

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree (S, FE) Examination June 2024 (2019 Scheme)

Course Code: CXT307**Course Name: VIRTUAL REALITY**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- | | | |
|----|--|---|
| 1 | Differentiate Virtual reality and Augmented reality. | 3 |
| 2 | Explain the concept of "tele-presence" in the context of Virtual Reality. | 3 |
| 3 | Explain the terms i) Virtual display, ii) Aural display. | 3 |
| 4 | Explain four key elements of virtual reality experience. | 3 |
| 5 | Describe the data structure used to encode the geometric models. | 3 |
| 6 | Describe the terms yaw, pitch, and roll in 3D rotation. | 3 |
| 7 | What is the primary cause of VR sickness, and how does it affect user's virtual reality experiences? | 3 |
| 8 | Explain the different phases of collision detection. | 3 |
| 9 | Write a note on teleportation in VR | 3 |
| 10 | Explain the terms
i) Remapping
ii) Manipulation in the context of virtual reality. | 3 |

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

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|----|----|--|----|
| 11 | a) | List out modern VR experiences in recent years. | 10 |
| | b) | Define the term interactivity in VR. | 4 |
| 12 | a) | Explain the evolution of VR systems. | 10 |
| | b) | List out the hardware components of a typical VR system. | 4 |

Module -2

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|----|----|--|----|
| 13 | a) | List out the different tracking methods in VR systems. | 10 |
| | b) | Write a note haptic display. | 4 |
| 14 | a) | Explain representation of virtual world. | 10 |
| | b) | List the user input devices in VR environment. | 4 |

Module -3

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|----|----|---|----|
| 15 | a) | Apply different matrices to transform a square face into all possible cases. | 14 |
| 16 | a) | Write a note on perception of motion in virtual reality | 9 |
| | b) | Consider a triangle with the coordinate values ((10, 10), (20, 10), (15, 15). Translate the triangle by $x_t = -8$ and $y_t = -7$. | 5 |

Module -4

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|----|----|--|---|
| 17 | a) | Apply Range-Kutta method to find an approximate value of y when x=0.2, where y=1 when x=0. (Hint: take value for h=0.1) for the equation $dy/dx=x+y$. | 8 |
| | b) | How do vestibular systems detect linear and angular motion? | 6 |
| 18 | a) | Explain the six types of vection in virtual environments, their characteristics, and the visual effects they produce. | 6 |

- b) How object motion is performed in the 3D world. 8

Module -5

- 19 a) Explain the concept of locomotion in the context of VR. 9
b) How can you implement social interaction in VR? Explain. 5
20 a) Explain how signal processing is employed for virtual reality. 10
b) How auralization is performed in Virtual Reality. 4
